How fast does sound travel?

Material	Speed of Sound (m/s)
Air at 0°C	330
Air at 20°C	334
Seawater	1, 540
Freshwater	1,410
Rock	2, 500
Steel	5, 000
Aluminium	5,100

In track events, the starter usually fires a gun that produces both a sound and a puff of smoke. The timers and spectators see the smoke of the gun first and hear the sound of the gun a short while later. This happens because light travels much faster than the sound from the gun. Light travels at 300, 000, 000 m/s, reaching the spectators with no noticeable delay. Sound travels through air at a 3little over 330 m/s. It takes a third of a second or more to reach the spectators, depending on how far they are from the starter. Sound travels quite a bit faster in solids and liquids than it does in air. The speed of sound in some materials is shown in the table.

Sound transmission in solids and liquids

Think about the way that the particles are arranged in solids, liquids and gases. Why is the speed of sound greater in solids and liquids than in air?

Some consequences of the speed of sound

Answer the following questions.

1. Why might the speed of sound be faster in saltwater than in freshwater?

2. In a mining operation, the sound from an underground explosion travels through the air and through the ground to a recording station 5 km away. Which s arrives first and by how much?



3. Though lightning and thunder occur together, we always see the lightning first and hear the thunder later. If a storm is 1 km away, how long does it take for the d to reach you?



4. In the old days a cowboy would listen for a train by putting his ear to the steel train tracks. If the train was 2 km away, how long would the sound take to reach him through the train tracks? How long would it take the sound to travel through the air?



5. After the sinking of the *Titanic* in 1912, many efforts were made to find ways of detecting icebergs in the dark or in fogs. By the 1950's, echo sounders for this purpose had become very reliable. Suppose a sound signal is sent from a ship, and reflects from an iceberg 1 km away. How long will this take if the sound travels through the water? How long would it take if the sound travels through the air?

